

GANESH GUPTA

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Owned end-to-end design of a RAG-based retrieval and ranking system using OpenAI's API, Pinecone vector search, and MCP tool-calling, integrating LLM capabilities deeply into production workflows at 99%+ uptime.
- Built an evaluation framework quantifying retrieval and generation quality, enforcing automated relevance, accuracy, and drift checks that blocked 15% of releases failing validation before reaching production.
- Ran experiments to improve retrieval relevance and accuracy, architecting AWS serverless infrastructure (Lambda, S3, DynamoDB, SageMaker) for real-time inference at sub-200ms latency.

QureAI (AI Startup)

Jan 2024 – May 2025

ML Infrastructure Engineer

- Diagnosed a search/retrieval quality gap directly from real usage signals, then owned the fix end-to-end: redesigned the embedding and chunking pipeline, lifting recall 61% to 89% via hybrid dense-sparse retrieval and re-ranking.
- Led migration of synchronous model serving to async vLLM with autoscaling as an early engineer at this AI startup, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices processing regulated financial transactions for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning, in a domain requiring auditable, traceable transaction records.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams through code reviews and design write-ups, reducing production escalations 60% in one quarter.

Projects

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL, built to keep agentic systems trustworthy at volume.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed, traceable root cause reports.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, SQL, Java, TypeScript

Search & Retrieval: RAG, Embeddings, Vector Databases (Pinecone, FAISS), Hybrid Dense-Sparse Retrieval, Re-Ranking, Evals

LLM & Agents: OpenAI API, Agentic AI, LangGraph, LangChain, OpenAI Agents SDK, MCP, Prompt Engineering, vLLM

Backend & Cloud: FastAPI, REST APIs, PostgreSQL, AWS (Lambda, EC2, S3, SageMaker), GCP (Cloud Run), Docker, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer